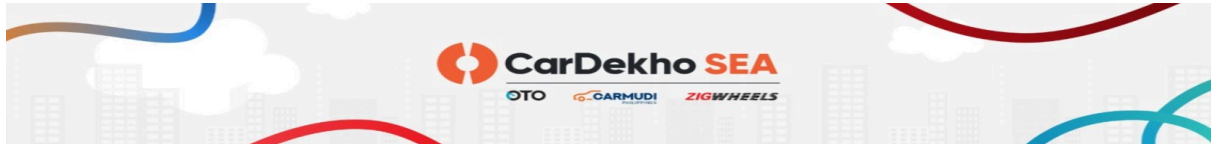


Modelling for Business Analytics

Analyzing the Impact of Lead Source on Conversion Rates and Response Time



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1. Problem Statement

In a competitive automotive marketplace like **CarDekho Southeast Asia**, understanding which lead sources generate the most valuable conversions is crucial. Dealers receive inquiries through multiple sources — *Affiliate, Dealer Referral, Facebook, Google Ads*, and *Organic search*. However, inconsistent follow-ups and delayed responses may affect conversion performance.

This study aims to **evaluate the impact of lead source and other operational factors (response time, lead score, dealer experience)** on conversion probability. The objective is to identify statistically significant drivers of lead conversion and suggest actionable improvements for sales efficiency.

2. Dataset Description

Variable	Description
LeadID	Unique identifier for each customer lead
DealerID	Unique dealer handling the lead
Region	Lead’s geographical location
CarMake / CarModel	Vehicle type of interest
LeadSource	Source: Google Ads, Facebook, Organic, Dealer Referral, Affiliate
ResponseTime_hrs	Time taken by dealer to contact lead
CarPrice_USD	Vehicle price
LeadScore	Internal CRM score (50–100)
FollowUps	Count of follow-ups done
DealerExperience_yrs	Dealer experience in years
CustomerType	First-time or repeat buyer
Converted	1 = Sale converted, 0 = Not converted

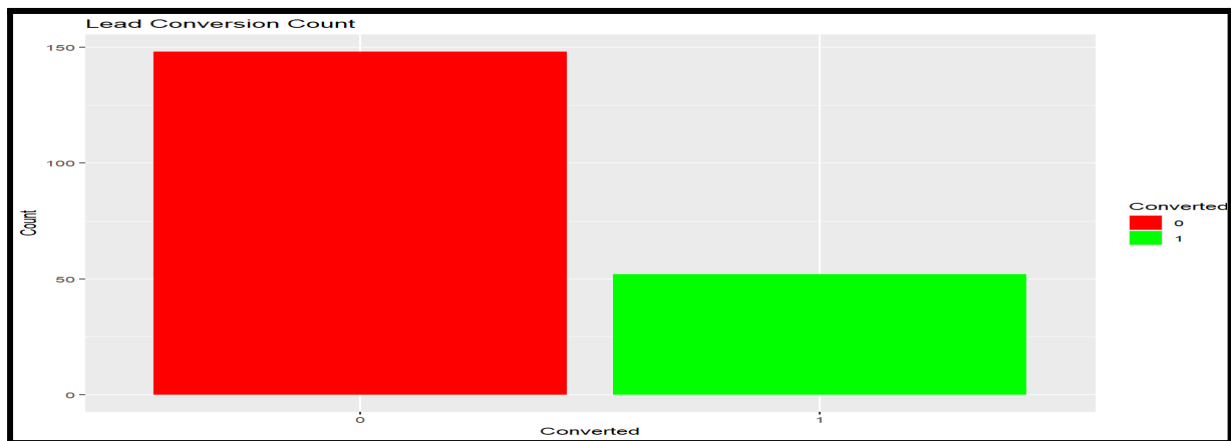
Tools Used: R (dplyr, ggplot2, car, caret, pscl, pROC)

3. Exploratory Data Analysis (EDA)

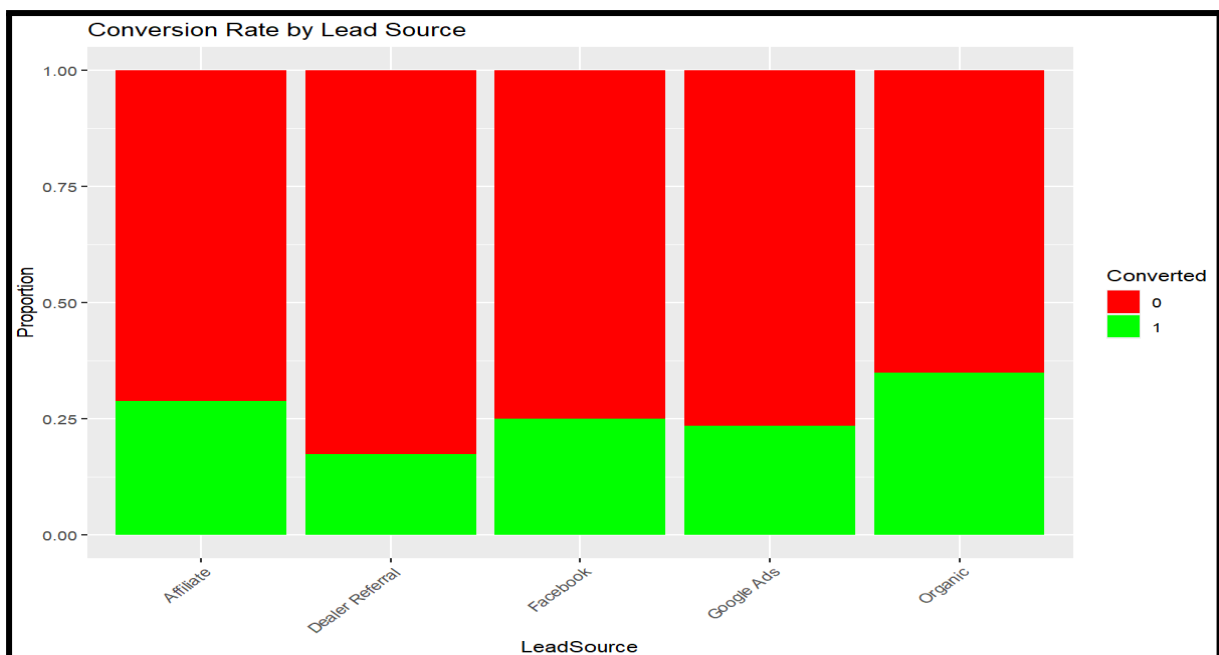
Conversion Rate Distribution

- Out of 200 leads, 26% converted (52 converted, 148 not converted).
- The majority of leads came from Google Ads and Dealer Referrals.
- Visuals:
 1. The bar plot of conversions showed imbalance (more 0s than 1s).
 2. Stacked bar (LeadSource vs Conversion) → roughly uniform conversion proportions across sources.
 3. Boxplot (Response Time vs Source) → median response time ~22 hours, no major visible outliers per source.

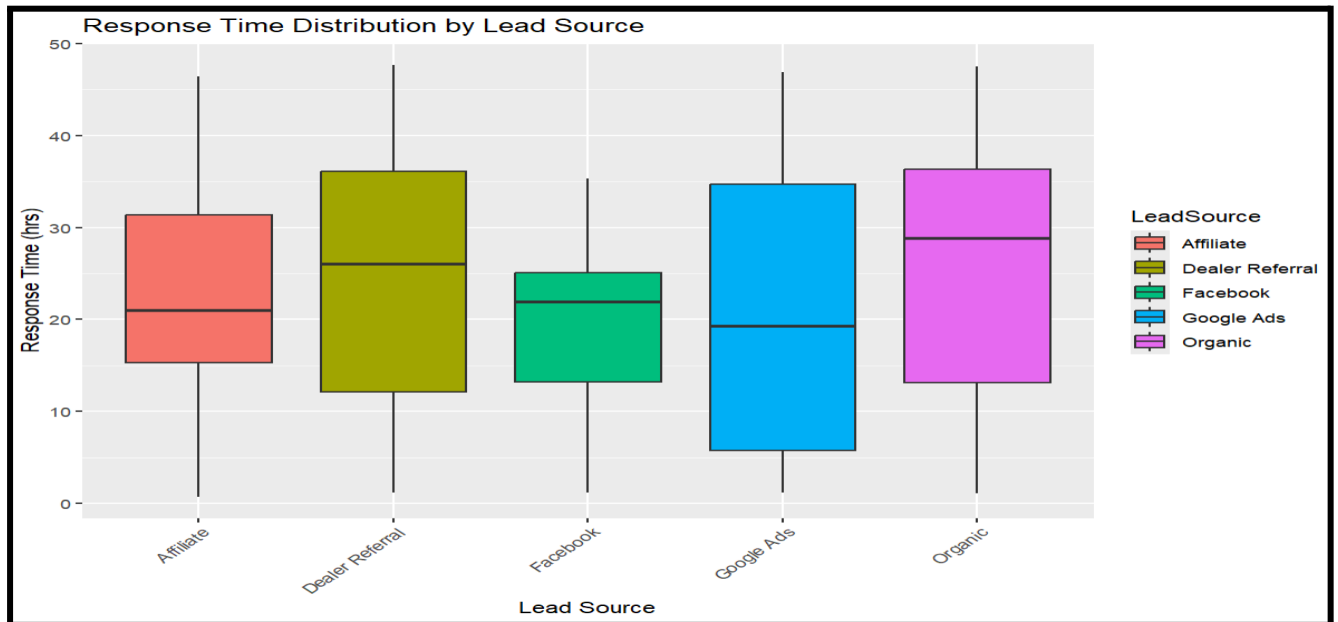
1. Lead Conversion Count - The bar chart below shows that conversion is low overall (~26%), indicating a high drop-off rate during follow-up.



2. LeadSource VS Conversion - Stacked proportional bars compared conversions across sources. Conversion ratios appeared similar across channels.



3. Response Time VS Source - Boxplots revealed moderate variation in response time, but no outliers suggesting any channel delay dominance.



4. Hypothesis Testing 1 – Effect of Lead Source on Conversion

Goal: Test if conversion rate differs across lead sources.

Hypotheses

- **H₀:** Conversion rate is the same across all lead sources.
- **H₁:** Conversion rate differs among lead sources.

Test Used: Chi-square test of independence

Statistic	Value
χ^2	3.855
df	4
p-value	0.4259

Decision: Fail to reject H₀ (since p > 0.05).

Interpretation: There is **no statistically significant difference** in conversion rate among different lead sources.

5. Linear Regression Analysis: Impact of Region and Lead Score on Response Time

Objective - The purpose of this analysis is to understand whether **response time** for customer leads differs significantly **across regions** and whether **lead quality (LeadScore)** influences how quickly dealers respond.

By modeling *ResponseTime_hrs* as a continuous dependent variable, this regression complements the logistic models, providing insight into **dealer behavior and operational efficiency**.

Model Specification :

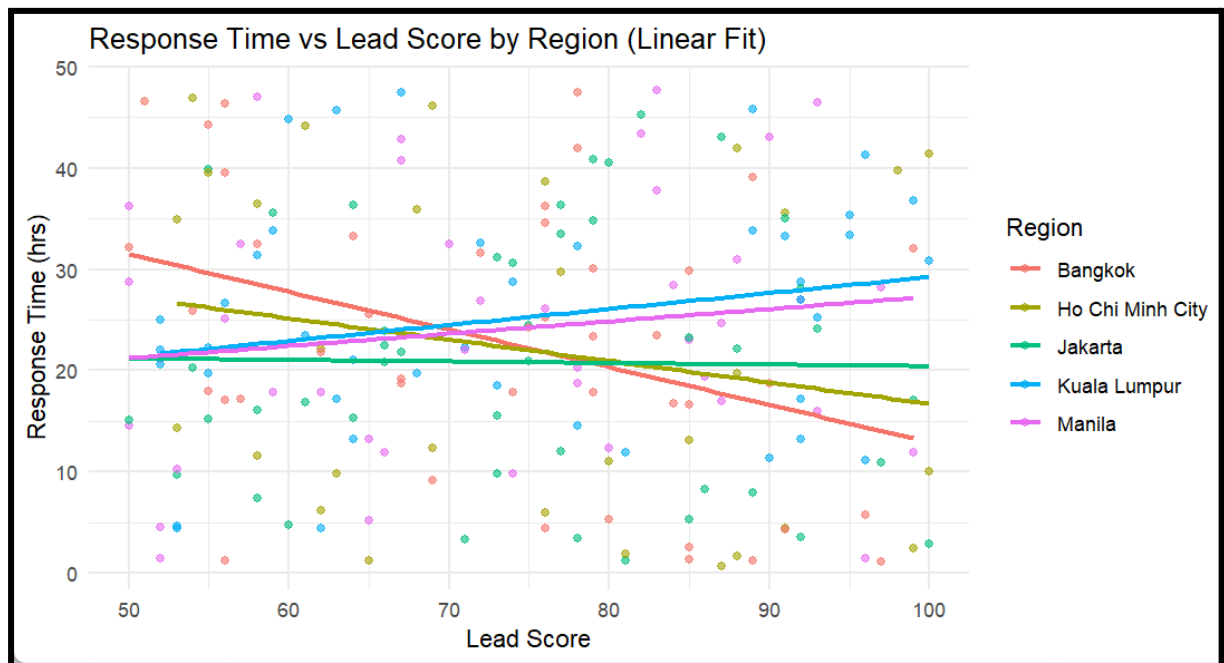
$$\text{ResponseTime_hrs} = \beta_0 + \beta_1(\text{Region}) + \beta_2(\text{LeadScore}) + \epsilon$$

Term	Estimate	Std. Error	t-value	p-value	Interpretation
(Intercept)	25.76	5.15	5.01	<0.001	Baseline response time is ~25.8 hours for the reference region.
Region (all)	Varies between -1.96 and +2.39	—	$p > 0.05$	No region shows a statistically significant deviation in response time.	
LeadScore	-0.04	0.064	-0.62	0.535	Lead quality does not significantly affect response time.

Interpretation

- The **intercept (25.76)** represents the **average dealer response time (~26 hours)** for the base region (Bangkok or first factor level).
- All **regional coefficients are statistically insignificant ($p > 0.05$)**, indicating that **response times do not differ meaningfully between regions**.
- The **LeadScore coefficient is negative (-0.04)** but not significant, implying a *slight trend* where higher-quality leads tend to receive faster responses — though the evidence is weak.
- The **overall model explains less than 2% of the variation ($R^2 = 0.016$)**, suggesting that **other factors (e.g., workload, dealer size, or lead channel)** might better explain response delays.

Response Time vs Lead Score by Region (Linear Fit) :The nearly flat slopes across regions visually confirm that LeadScore has minimal impact on response time and that regional differences are negligible.



6. Logistic Regression 1 – Conversion ~ Lead Source

Objective: Estimate the likelihood of conversion as a function of lead source only.

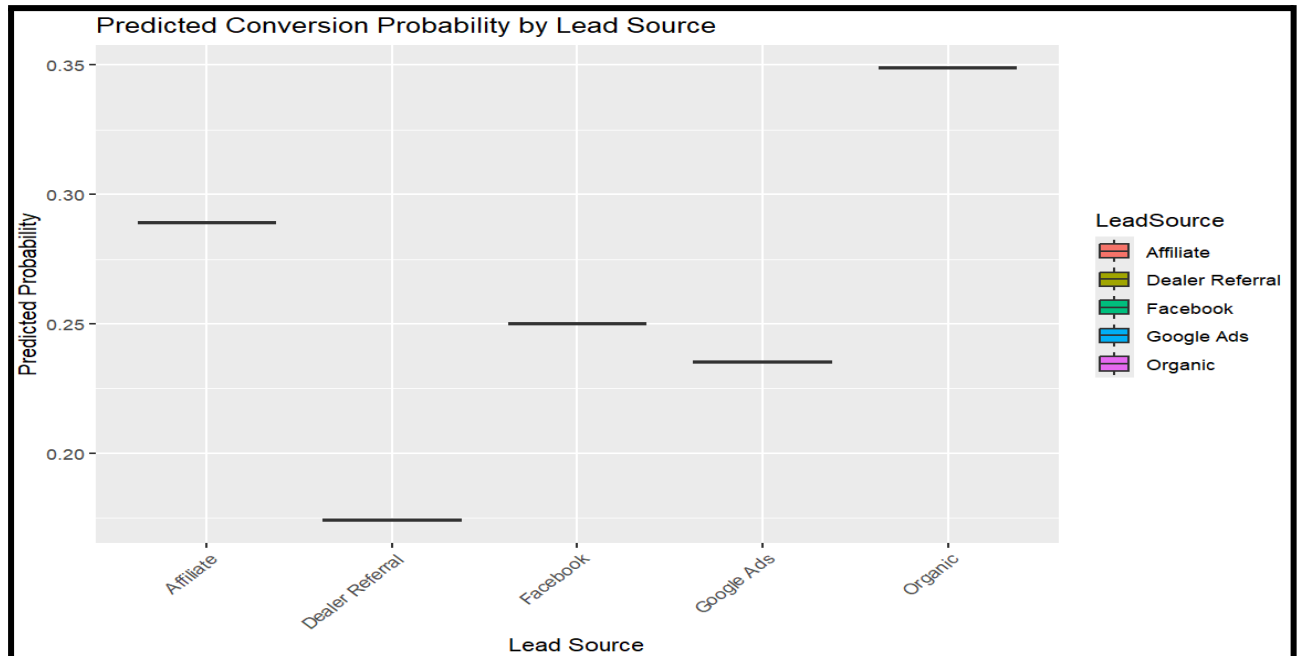
Model: `glm(Converted ~ LeadSource, family = binomial)`

Model Summary

Predictor	Estimate	p-value	Odds Ratio	Interpretation
Intercept (Affiliate baseline)	-0.9008	0.006	0.41	Affiliate leads have ~41% baseline odds of conversion
Dealer Referral	-0.6574	0.197	0.52	Slightly lower odds, not significant
Facebook	-0.1978	0.706	0.82	Similar odds to baseline
Google Ads	-0.2779	0.594	0.76	No significant difference
Organic	0.2766	0.547	1.32	Slightly higher odds, not significant

Interpretation: None of the sources show a significant difference ($p > 0.05$). The baseline (Affiliate) has an estimated conversion odds of 0.41 (41%), and other sources are similar in effectiveness.

Predicted Conversion Probability by Lead Source - *Organic and Affiliate* sources show slightly higher medians, consistent with odds ratios.



7. Logistic Regression 2 – Enhanced Model (Full Predictors)

Objective: To include operational and quality predictors to isolate the true source impact.

Model: $\text{Converted} \sim \text{LeadSource} + \text{ResponseTime_hrs} + \text{LeadScore} + \text{FollowUps} + \text{DealerExperience_yrs} + \text{CarPrice_USD}$

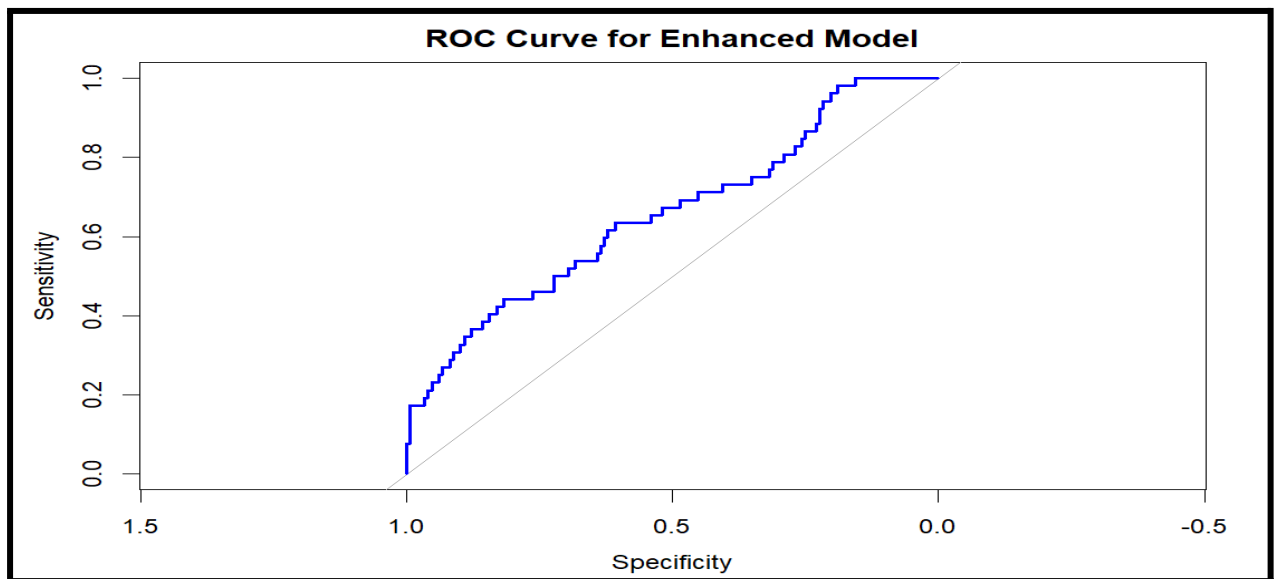
Predictor	Estimate	p-value	Odds Ratio	Interpretation
ResponseTime_hrs	-0.0410	0.0027	0.96	Significant negative impact: every hour of delay reduces odds by ~4%
LeadScore	0.015	0.196	1.02	Positive effect, not significant
LeadSource (all)	—	> 0.05	—	Still not significant
DealerExperience_yrs	-0.036	0.549	0.96	No effect
FollowUps	-0.016	0.841	0.98	No effect
CarPrice_USD	-0.000028	0.559	1.00	No effect

Model Fit Metrics:

- McFadden's $R^2 = 0.072$
- Accuracy = 78%
- AUC = 0.657 (moderate discrimination)
- Sensitivity = 0.17, Specificity = 0.99

Interpretation: Response time is the **only significant predictor** of conversion. Slower responses drastically reduce conversion odds. Lead source, pricing, and dealer experience show no measurable influence.

ROC Curve for Enhanced Logistic Model



8. Hypothesis Testing 2 – Lead Source vs Response Time (ANOVA)

Goal: Test whether the average response time differs across lead sources.

Hypotheses

- H_0 : Mean response time is equal across all lead sources.
- H_1 : At least one lead source differs.

Result

Statistic	Value
F(4,195)	1.619
p-value	0.171

Decision: Fail to reject H_0 ($p > 0.05$).

Interpretation: Mean response time is **not significantly different** across lead sources.

Tukey's post-hoc test also confirmed no pairwise differences (all adjusted p-values > 0.26).

```
> TukeyHSD(anova_source)
Tukey multiple comparisons of means
 95% family-wise confidence level

Fit: aov(formula = ResponseTime_hrs ~ LeadSource, data = lead_data)

$LeadSource
              diff      lwr      upr      p adj
Dealer Referral-Affiliate  2.815604 -4.817504 10.448711 0.8479371
Facebook-Affiliate        -2.154097 -10.572505  6.264311 0.9552647
Google Ads-Affiliate      -1.582222 -9.854663  6.690219 0.9845493
Organic-Affiliate         4.020103  -3.743560 11.783767 0.6118128
Facebook-Dealer Referral  -4.969701 -13.349995  3.410593 0.4782755
Google Ads-Dealer Referral -4.397826 -12.631477  3.835825 0.5827318
Organic-Dealer Referral    1.204499  -6.517819  8.926818 0.9928631
Google Ads-Facebook        0.571875  -8.394634  9.538384 0.9997855
Organic-Facebook          6.174201  -2.325180 14.673581 0.2697774
Organic-Google Ads        5.602326  -2.752503 13.957154 0.3501405
```

9. Model Evaluation

Goal: Check model fit and predictive performance.

- $R^2 = 0.072 \rightarrow$ model explains $\sim 7\%$ of variability (typical for logistic models).
- **Confusion Matrix:**
 - Accuracy = 78%
 - Sensitivity = 17% (misses many positives)
 - Specificity = 99% (correctly identifies non-conversions).
- **ROC Curve & AUC = 0.657** $\rightarrow$ Fair model performance.
- Threshold = 0.34 gives best balance of sensitivity & specificity.

The model has fair predictive ability, but limited power — mostly due to imbalance.

10. Summary of Results

Goal: Check model fit and predictive performance.

Test / Model	Variable of Interest	p-value	Conclusion
Chi-Square	LeadSource (on Conversion)	0.426	No significant effect
Logistic Regression	ResponseTime_hrs	0.0027	Significant negative impact
ANOVA	LeadSource (on ResponseTime)	0.171	No significant difference

Key Finding:

Lead Source has **no direct effect** on conversion, but **Response Time** significantly reduces conversion odds when delayed.

11. Business Insights & Recommendations

Insight	Actionable Recommendation
Response time is critical.	Implement automated dealer alerts to ensure responses within 12 hours.
Lead source not significant.	Maintain balanced marketing spend; focus on internal response process optimization.
Moderate model accuracy (78%).	Use a logistic model as a lead prioritization tool — high LeadScore + low response time = high conversion probability.
AUC = 0.657 (moderate).	The model can be refined using more behavioral variables (call logs, dealer ratings).

12. Conclusion

The study reveals that **conversion success is process-driven, not source-driven**.

While different channels (Google Ads, Facebook, etc.) perform similarly, the **speed of dealer response** plays a decisive role.

Each additional hour of delay reduces conversion odds by approximately **4%**, making responsiveness the key operational metric.

In summary:

“It’s not where the lead comes from — it’s how fast you act that matters.”